IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

CONTROL SYSTEMS

Wednesday, 29 April 2026, 2:30 PM

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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CONTROL SYSTEMS – 2025/2026

Information for candidates:

Remark for the Students: the information provided here is much more than needed to answer the Exam Paper Questions; that is, the presence of a specific formula does not necessarily mean that this formula is required to answer the questions.

- **Bode Plots to answer Question 3**

Remark for the Students: In case drawing Bode plots is deemed helpful or necessary to answer Question 3), an additional answer book is provided that contains a few clean semi-logarithmic charts that should be used and should be attached to the main Answer Book.

- **Miscellaneous**

– *Settling-time* at 1% for asymptotically stable first-order systems: $t_{s,0.01} \simeq 4.6 \cdot \tau$ where τ denotes the time-constant (that is, the inverse of the pole with a sign-change).

– *Settling-time* at 1% for asymptotically stable second-order systems with complex-conjugate poles: $t_{s,0.01} \simeq \dfrac{4.6}{\xi\,\omega_n}$ where ω_n denotes the natural angular frequency and ξ the damping-ratio.

– *Realisation of a transfer function*: given the strictly proper transfer function

$$G(s) = \frac{\beta_m s^m + \beta_{m-1} s^{m-1} + \cdots + \beta_1 s + \beta_0}{s^n + \alpha_{n-1} s^{n-1} + \cdots + \alpha_1 s + \alpha_0} \quad \text{with} \quad m < n$$

a possible state-space realisation is

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -\alpha_0 \\ 1 & 0 & \cdots & 0 & -\alpha_1 \\ \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 & -\alpha_{n-2} \\ 0 & \cdots & 0 & 1 & -\alpha_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \\ 0 \\ \vdots \\ 0 \end{bmatrix} u \\ \\ y = \begin{bmatrix} 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \end{cases}$$

Information for candidates (continued):

Main Laplace Transforms and General Properties

$f(t)$	$F(s)$
$1(t)$	$\frac{1}{s}$
$t \cdot 1(t)$	$\frac{1}{s^2}$
$t^2 \cdot 1(t)$	$\frac{2}{s^3}$
$e^{\alpha t} \cdot 1(t)$	$\frac{1}{s-\alpha}$
$t \cdot e^{\alpha t} \cdot 1(t)$	$\frac{1}{(s-\alpha)^2}$
$\sin(\omega t) \cdot 1(t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t) \cdot 1(t)$	$\frac{s}{s^2+\omega^2}$
$t \cdot \sin(\omega t) \cdot 1(t)$	$\frac{2\omega s}{(s^2+\omega^2)^2}$
$t \cdot \cos(\omega t) \cdot 1(t)$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
$e^{\sigma t} \cdot \sin(\omega t) \cdot 1(t)$	$\frac{\omega}{(s-\sigma)^2+\omega^2}$
$e^{\sigma t} \cdot \cos(\omega t) \cdot 1(t)$	$\frac{s-\sigma}{(s-\sigma)^2+\omega^2}$
$t \cdot e^{\sigma t} \cdot \sin(\omega t) \cdot 1(t)$	$\frac{2\omega(s-\sigma)}{\left[(s-\sigma)^2+\omega^2\right]^2}$
$t \cdot e^{\sigma t} \cdot \cos(\omega t) \cdot 1(t)$	$\frac{(s-\sigma)^2-\omega^2}{\left[(s-\sigma)^2+\omega^2\right]^2}$

Table 1 Time-functions and corresponding Laplace transforms

$\mathscr{L}[f(t)+g(t)]$	$F(s)+G(s)$
$\mathscr{L}\left[\int_0^t f(\tau)d\tau\right]$	$\frac{1}{s}F(s)$
$\mathscr{L}\left[\dot{f}(t)\right] = \mathscr{L}\left[\frac{d}{dt}f(t)\right]$	$sF(s)-f(0)$
$\mathscr{L}\left[\ddot{f}(t)\right] = \mathscr{L}\left[\frac{d^2}{dt^2}f(t)\right]$	$s^2F(s)-sf(0)-\dot{f}(0)$
$\mathscr{L}\left[\frac{d^3}{dt^3}f(t)\right]$	$s^3F(s)-s^2f(0) -s\dot{f}(0)-\ddot{f}(0)$
$\mathscr{L}\left[e^{kt}f(t)\right]$	$F(s-k)$
$\mathscr{L}[f(t-t_0)]$	$e^{-st_0}F(s)$
$\mathscr{L}[f(at)]\,, a \in \Re, a>0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
$\mathscr{L}[t^n f(t)]$	$(-1)^n\frac{d^n}{ds^n}F(s)$
$\mathscr{L}[f(t)*g(t)] =$	
$= \mathscr{L}\left[\int_0^t f(\tau)g(t-\tau)d\tau\right]$	$F(s)G(s)$
If $\deg[N(s)] < \deg[D(s)]$ with $F(s)=N(s)/D(s)$	$f(0)=\lim_{s\to\infty}F(s)$
If all poles of $F(s)$ have negative real part except (possibly) one pole in $s=0$	$\lim_{t\to\infty}f(t)=\lim_{s\to 0}sF(s)$

Table 2 Main properties of Laplace tranforms

1. Consider the block scheme shown in Figure 1.1:

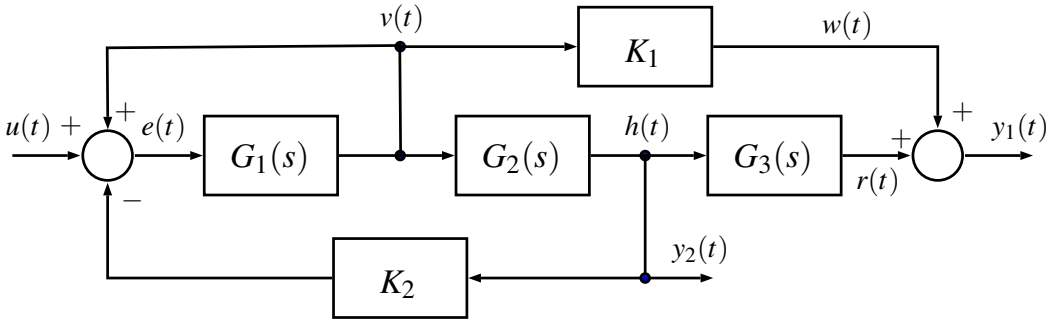


Figure 1.1 Block scheme for Question 1.

where, with some abuse of notation, we represent the inputs and outputs of each block in the time-domain instead of in the Laplace domain. The transfer functions $G_1(s)$, $G_2(s)$, and $G_3(s)$ are strictly proper and have no hidden common factors in their numerators and denominators. Moreover, $K_1, K_2 \in \mathbb{R}$ are scalars.

a) Determine the transfer function $G_{u,y_1}(s)$ from the input $u(t)$ and the output $y_1(t)$ when $K_1 \neq 0$ and $K_2 \neq 0$.

[10 Marks]

b) Determine the transfer function $G_{u,y_2}(s)$ from the input $u(t)$ and the output $y_2(t)$ when $K_1 \neq 0$ and $K_2 \neq 0$.

[5 Marks]

c) Determine the transfer function $G_{u,y_1}(s)$ from the input $u(t)$ and the output $y_1(t)$ when $K_1 = 0$ and $K_2 = 0$.

[3 Marks]

d) Setting $K_1 = 0$ and

$$G_1(s) = \frac{1}{s+1}; \quad G_2(s) = \frac{1}{s-1}; \quad G_3(s) = \frac{1}{s+2}$$

i) Devise a state-space description of the dynamic system shown in Figure 1.1 of the form:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

where $x \in \mathbb{R}^n$ is a state vector to be suitably defined, $\dot{x}$ denotes its time-derivative, u denotes the input vector, y denotes the output vector, and A, B, C, D are suitable matrices to be determined.

[12 Marks]

ii) Based on the state equations obtained in your answer to Question 1-d)-i), establish the stability properties of the dynamic system shown in Figure 1.1 as a function of the scalar $K_2 \in \mathbb{R}$.

[5 Marks]

iii) Plug $K_1 = 0$ and the specific transfer functions $G_1(s)$, $G_2(s)$, and $G_3(s)$ into the transfer function $G_{u,y_1}(s)$ obtained in your answer to Question 1-a). Show that the stability properties established in your answer to Question 1-d)-ii) can be obtained also by analysing this specific transfer function $G_{u,y_1}(s)$.

[5 Marks]

2. Consider the linear time-invariant dynamic system described by the block scheme shown in Figure 2.1:

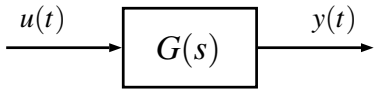


Figure 2.1 Block scheme for Question 2.

where, with some abuse of notation, we represent the input and the output in the time-domain instead of in the Laplace domain. It is assumed that the transfer function $G(s)$, has no hidden common factors in its numerator and denominator and is given by:

$$G(s) = 4 \cdot \frac{3e^{-s}}{9 + 6s + s^2}$$

a) Set

$$u(t) = [\sin(3t)] \cdot \mathbf{1}(t)$$

where $\mathbf{1}(t)$ denotes the unit step function.

i) Show that there exists a sinusoidal regime (e.g., steady-state) output function $y_{\text{reg}}(t), t \in \mathbb{R}$ without determining $y_{\text{reg}}(t)$.

[4 Marks]

ii) Determine the analytical expression of the output function regime (e.g., steady-state) output function $y_{\text{reg}}(t), t \in \mathbb{R}$.

[5 Marks]

b) Set the input as

$$u(t) = \left(1 - e^{-t}\right) \mathbf{1}(t) - \left[1 - e^{-(t-7)}\right] \mathbf{1}(t-7)$$

where $\mathbf{1}(t)$ denotes the unit step function.

i) Determine the analytical expression of the output function $y(t), t \in \mathbb{R}$.

[8 Marks]

ii) Sketch the diagrams of the time-behaviours of the input $u(t), t \in \mathbb{R}$ and the output $y(t), t \in \mathbb{R}$.

[3 Marks]

3. Consider the closed-loop control system represented by the block scheme shown in Figure 3.1:

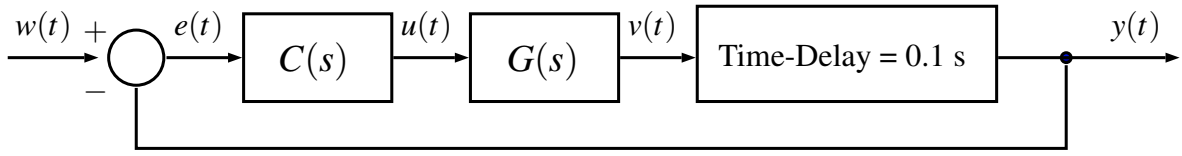


Figure 3.1 Block scheme for Question 3.

where, with some abuse of notation, we represent the inputs and outputs of each block in the time-domain rather than the Laplace domain. Moreover

$$G(s) = 12 \cdot \frac{1+0.1s}{1+24s}$$

a) Consider a controller $C(s)$ with a fixed structure of the form

$$C(s) = \mu_C \frac{1+s\tau}{s}$$

where $\mu_C > 0$ and $\tau > 0$ are two parameters to be designed.

Determine $\mu_C > 0$ and $\tau > 0$ such that:

i) The closed-loop steady-state error $\bar{e}_\infty$ satisfies:

$$w(t) = A \cdot t \cdot \mathbf{1}(t)\,,\ \forall A \text{ such that } |A| \leq 1 \quad \Longrightarrow \quad |\bar{e}_\infty| \leq 0.5$$

where $\mathbf{1}(t)$ denotes the unit step function.

ii) The phase margin φ_m satisfies $\varphi_m \geq 60^\circ$.

iii) The settling-time at 1% satisfies $t_{s,0.01} \leq 2\,\text{s}$.

If needed or deemed useful, use the semi-logarithmic charts provided in the Additional Answer Book.

[28 Marks]

b) With reference again to the closed-loop control system represented by the block scheme shown in Figure 3.1, now consider a **different controller** $\widetilde{C}(s)$ given by

$$\widetilde{C}(s) = 1$$

i) Establish whether the closed-loop system under the action of the new controller $\widetilde{C}(s)$ is asymptotically stable.

[7 Marks]

ii) Establish whether the Bode criterion is applicable to the closed-loop system under the action of the **new controller** $\widetilde{C}(s)$ to ascertain the closed-loop stability properties.

[5 Marks]

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ADDITIONAL ANSWER BOOK FOR BODE PLOTS

PLEASE WRITE HERE YOUR STUDENT CID $\Longrightarrow$

Information for candidates:

This additional answer book provides semi-logarithmic charts to be used to sketch Bode plots (if deemed helpful or necessary to answer Question 3) and should be attached to the main Answer Book.

